# TP05 – Apache Kafka

## Report Questions and Answers

Real-Time Data Streaming and Processing

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| **Module** | Master 2 – Artificial Intelligence & Data Science |
| **University** | University of El Oued |
| **Academic Year** | 2026/2027 |

Prepared report containing short and simple answers to the nine TP05 questions.

# Report Questions

The answers below use simple, concise English and follow the TP05 concepts.

## 1. What problem does Kafka solve?

Kafka allows applications to send, receive, and process large amounts of data in real time. It helps different systems communicate and exchange data efficiently.

## 2. Define Producer, Consumer, Broker, Topic, Partition, Offset, and Consumer Group.

**Producer:** Sends messages to Kafka.
**Consumer:** Reads messages from Kafka.
**Broker:** A Kafka server that stores and manages messages.
**Topic:** A category where messages are stored.
**Partition:** A part of a topic that stores messages in order.
**Offset:** The position of a message inside a partition.
**Consumer Group:** A group of consumers that work together to read messages from a topic.

## 3. What is the difference between a Topic and a Partition?

A topic is a category for storing messages. A partition divides a topic into smaller parts to improve data distribution and parallel processing.

## 4. What does auto_offset_reset="earliest" mean?

It means that when a consumer has no previous offset, it starts reading messages from the beginning of the topic.

## 5. What is the purpose of group_id?

The group_id identifies a consumer group. Consumers with the same group ID share the work of reading messages from a topic.

## 6. Why did we use JSON? What is one limitation?

I used JSON because it is simple, readable, and easy to exchange between different applications. One limitation is that JSON can be larger than binary formats.

## 7. What is the benefit of Docker Compose?

Docker Compose makes it easier to configure and run Kafka using one configuration file. It also simplifies starting and stopping the service.

## 8. Why is a single Kafka broker not enough for production?

A single broker is a single point of failure. If it stops working, the service becomes unavailable. Production systems need multiple brokers for better availability and reliability.

## 9. What is the role of partitions in Kafka?

Partitions allow Kafka to distribute messages and process them in parallel. More partitions can improve performance when multiple consumers are working in the same consumer group.